



Out-of-Band (OoB) Network Management

So far in all the lab exercises, you have been using the console cables to directly connect to the console ports to configure and manage the network devices.

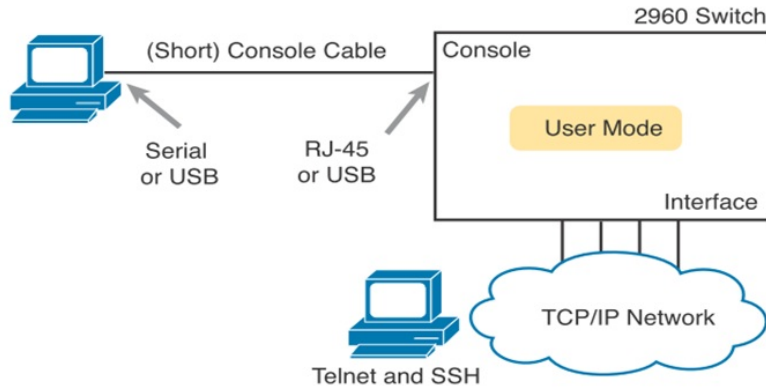


Figure 1

In fact as shown in Figure 1, it is possible to use **Telnet** or **SSH** over the **TCP/IP network** to **remotely connect** to manage the network devices which is more convenient and is commonly implemented in practice.

However, remotely managing the network devices via **in-band management** over the enterprise LAN (aka **main production network**) is not ideal since it can be subjected to **attacks** by malicious users. To be secure, it is recommended to set up a separate **management LAN** which is isolated from the main production network for remote management of the network devices called **out-of-band (OoB)** management as shown in Figure 2.

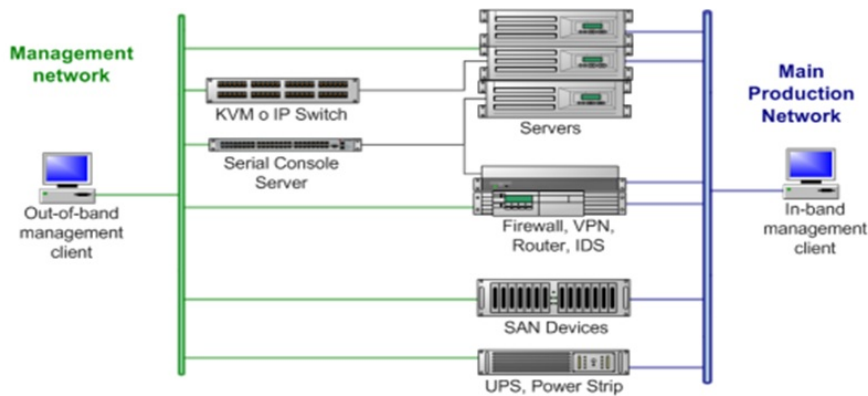


Figure 2

To support the setup of **management LAN**, network devices are commonly manufactured with **management ports** which are typically located next to the console ports. Examples of management (mgmt) ports on Cisco Catalyst 9200L switch, 9300L switch, and Firepower 1010 firewall are shown in Figure 3, 4 and 5 respectively.



Figure 3



Figure 4



Figure 5

Virtual Routing and Forwarding (VRF)

You may have noticed that Cisco Catalyst 8200L router does not have a management port. Nevertheless, it supports **virtual routing and forwarding (VRF)** which can provide **virtual separation** between **management LAN** and main **production network** so that network traffic will not cross over between them. In fact, management ports may be implemented using VRF, hence called management VRF.

Essentially, **VRF** creates **different routing tables** so that network traffic within one routing table will not cross over to mix with network traffic belonging to other routing tables, which conceptually is logically dividing a router into separate routers.

To have a better understanding of **VRF**, let's take a look at an example of implementing VRF on a 5-port router as shown in Figure 6.

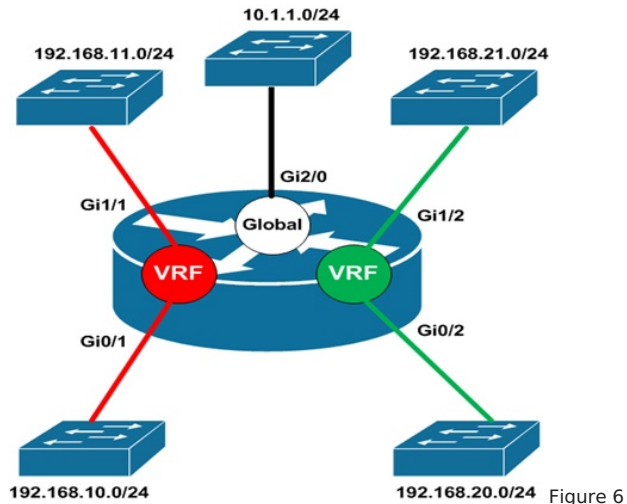


Figure 6

Suppose that ports Gi0/1 and Gi1/1 are connected to the **management LAN** to be configured to **VRF Red**. Port Gi2/0 is connected to the production LAN but will not be configured with VRF. Ports Gi0/2 and Gi1/2 are connected to another separate LAN to be configured to VRF Green.

Using Cisco Catalyst 8200 series router as an example, VRF can be configured by first creating a VRF instance using the command:

```
router(config)# vrf definition vrf-name
router(config-vrf)# address-family ipv4
```

Next, associate an interface to a VRF instance by configuring:

```
router(config)# interface interface-id
router(config-if)# vrf forwarding vrf-name
router(config-if)# ip address ip-addr subnet-mask
```

After configuring VRF, notice that the default or global routing table will only have the route for the port not configured to VRF as shown in Figure 7, thus isolating production network traffic from other VRFs.

```
Router# show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    10.1.1.0/24 is directly connected, GigabitEthernet2/0
```

Figure 7

To display the separate routing table belonging to a VRF instance, enter the command:

```
router# show ip route vrf vrf-name
```

For example, Figure 8 shows the routing table for VRF Red which only contains routes within the management LAN, thus isolating management traffic from the other LANs.

```

Router# show ip route vrf Red
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.10.0/24 is directly connected, GigabitEthernet0/1
C    192.168.11.0/24 is directly connected, GigabitEthernet1/1

```

Figure 8

Jump Host, Jump Server, or Jump Box

In practice, it is desirable to extend remote management of network devices from the Internet. However, it is too risky to simply expose the management LAN for all the network devices to be directly accessible. Instead, a **jump host** or **jump server** or **jump box** is commonly implemented as shown in Figure 9.

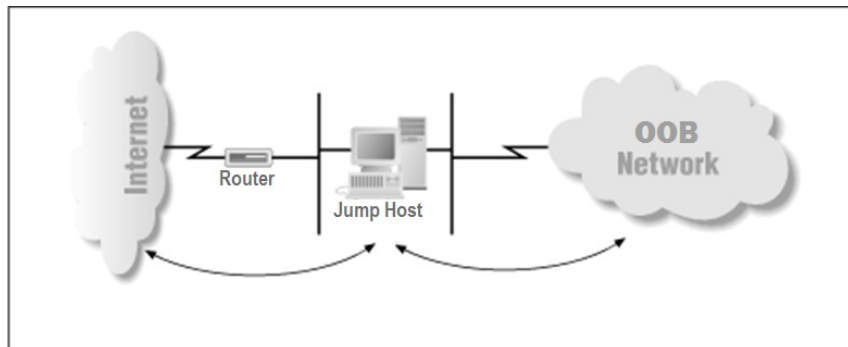


Figure 9

Essentially, a **jump host** is an intermediary device that acts as an entry point from a **less-secure zone** to a **more-secure zone**. With a jump host, the attack surface is reduced to a single point of entry, which will be easier to log, monitor and audit.

However, the drawback is that jump host can become an attractive target to attacks, and be the single point of failure. Thus, securely-hardened jump host is a must. A typical implementation is to have a hardened Linux machine configured with SSH hosted at the DMZ which is then permitted to access the OoB management LAN.

There are numerous guides from the Internet on the setting up and accessing an SSH jump host. Some examples are:

[How to Access a Remote Server Using a SSH Jump Host](#)

[How to Setup SSH Passwordless Login in Linux](#)

[How to Secure and Harden OpenSSH Server](#)

You may be interested to also read [Inside the Cyberattack That Shocked the US Government](#) due to the compromise of its jumpbox.

Secure Shell (SSH)

Besides setting up the management LAN and jump host, the **network devices** will also need to be enabled with **SSH** in order to be managed remotely. (Telnet is not recommended as it is insecure and hence will not be covered.)

First, configure hostname, e.g. R1, and domain name to be used for key generation:

```

router(config)# hostname host-name
R1(config)# ip domain-name domain-name

```

Second, generate an RSA key-pair to be used by SSH, where *modulus-size* is typically 2048 bits or 4096 bits to be more secure:

```

R1(config)# crypto key generate rsa modulus modulus-size

```

Next, enable SSH version 2 which is the current recommended standard:

```

R1(config)# ip ssh version 2

```

Optionally, configure number of authentication retries if login is not successful, and timeout for inactivity:

```

R1(config)# ip ssh authentication-retries number-of-retries
R1(config)# ip ssh timeout timeout-in-seconds

```

To verify SSH status, you may use the commands:

```

R1(config)# show ip ssh
R1(config)# show ssh

```

Now, the network devices are ready for **remote management**. To control access to the network devices, **authentication**, **authorization** and **accounting (AAA)** will need to be implemented which will be covered in the next few lab exercises.